

PROJECT PROPOSAL

Deep Neural Networks — Research Paper Project

Trustworthy Deep Learning for Chest X-ray Disease Detection:

Benchmarking the Robustness, Explainability and Calibration of CNNs and Vision Transformers under Dataset Shortcut Bias

A comparative study that goes beyond accuracy — asking not only which model is most accurate, but which is most trustworthy when the data itself is biased.

Project type	Comparative / benchmarking research study (no new architecture required)
Target output	Conference or undergraduate journal paper
Team size	5 members
Duration	
Primary framework	Python · PyTorch · timm · Grad-CAM · SHAP
Date	July 2026

Contents

1. Executive Summary

Chest X-ray (CXR) is the most common medical imaging examination in the world, and deep learning models can now classify chest diseases with reported accuracies above 95%. However, a large and growing body of work shows that many of these high accuracy numbers are misleading: models frequently succeed by exploiting hidden dataset artefacts — such as which hospital or source a scan came from — rather than by reading the lungs. When these models are tested on data from a new hospital, their performance collapses. This is called shortcut learning, and it is the single biggest obstacle to deploying CXR AI in real clinics.

This project does not attempt to invent a new neural network. Instead, it delivers a rigorous, reproducible comparative study of four widely used transfer-learning architectures — three CNNs (ResNet50, DenseNet121, EfficientNet-B0) and one Vision Transformer (ViT-Base) — evaluated under a single unified protocol on the same data. Crucially, the study is framed around trustworthiness, not just accuracy. We ask four questions that the current literature answers only in fragments: (1) which architecture is most accurate, (2) which is best calibrated (i.e. knows when it is uncertain), (3) whose explanations most faithfully point at the diseased lung region rather than at image artefacts, and (4) which architecture generalises best to an unseen external dataset.

Why this framing wins. Our literature review (Section 3) found dozens of 2024–2025 papers that simply compare CNN vs Vision Transformer accuracy on chest X-rays — this angle is saturated and reviewers treat it as incremental. The genuine, defensible gap is that almost no study combines accuracy, calibration, explanation faithfulness, computational efficiency, and cross-dataset robustness into one controlled comparison, and even fewer use explainability as a diagnostic tool to expose shortcut learning on the popular COVID-19 Radiography dataset. That combination is our contribution and it is achievable in one month.

2. Problem Statement and Motivation

Chest diseases such as pneumonia, COVID-19, tuberculosis and lung opacity are among the leading causes of death worldwide. Radiologists diagnose them from chest X-rays, but interpretation is time-consuming, depends heavily on experience, and many hospitals — particularly in low-resource settings — have too few radiologists for the volume of scans requiring rapid screening. AI-assisted triage is therefore highly attractive.

The problem is not building an accurate classifier — that is now easy. The problem is trust. A CXR model intended for clinical support must satisfy three conditions that raw accuracy does not capture:

- **It must look at the right thing.** The model's decision should be driven by the pathological region of the lung, not by text markers, borders, or acquisition-device signatures.
- **It must know when it is unsure.** An over-confident wrong prediction is dangerous; the model's confidence scores must be calibrated so clinicians can trust a 90% output to be right roughly 90% of the time.
- **It must generalise.** Performance must hold up on images from hospitals and machines never seen during training.

This project builds and evaluates an explainable, deployment-aware CXR classification framework that measures all three of these properties across CNN and Transformer architectures, rather than reporting accuracy alone.

3. Research Gap Analysis

Before committing to a design we surveyed recent (2023–2026) literature. The findings below shape the entire proposal and are the reason the study is framed around robustness rather than a plain accuracy race.

3.1 What has already been done (and is therefore not novel)

- **Plain CNN-vs-ViT accuracy comparisons are saturated.** Numerous 2024–2025 papers already benchmark ResNet/EfficientNet/DenseNet against Vision Transformers on chest X-rays and report which is more accurate. A straightforward re-run of this comparison offers little new and is likely to be rejected as incremental.
- **Single-property studies exist in isolation.** There are separate papers on Grad-CAM explainability for CXR, on temperature-scaling calibration for medical images, and on efficiency benchmarking — but they rarely appear together under one controlled protocol.
- **Hybrid CNN-Transformer architectures are trending.** Recent work (e.g. ConvFormer/CaFormer-style hybrids, class-attention pooling ViTs) chases marginal AUROC gains. Competing on raw AUROC against these is not realistic for a one-month undergraduate project.

3.2 The documented problem the field keeps hitting

Multiple independent studies show that the popular publicly-mixed COVID-19 chest X-ray datasets suffer from source bias / shortcut learning: because the COVID-positive, pneumonia and normal images were collected from different source databases and hospitals, models learn to recognise the source rather than the disease. Reported internal AUCs above 0.99 have been shown to fall to ~0.76 on external data. Lung-segmentation and feature-disentanglement studies confirm that removing background artefacts changes both accuracy and where the model looks.

The research gap we exploit

No existing study systematically compares CNNs and Vision Transformers on chest X-rays across ALL of: accuracy, confidence calibration (ECE / reliability), explanation faithfulness, computational efficiency, AND cross-dataset robustness — under one identical training and preprocessing protocol.

In particular, it is unknown whether the global-attention inductive bias of Vision Transformers makes them MORE or LESS prone to shortcut learning than convolutional models on the biased COVID-19 Radiography dataset, and whether their attention maps reveal that bias more clearly than CNN Grad-CAM.

We use explainability not merely as a visualisation, but as an audit tool to detect and quantify shortcut learning — a fresh and clinically meaningful angle.

3.3 Our positioning

The paper's honest, reviewer-proof claim is therefore: a unified, reproducible benchmark of four transfer-learning models that ranks them not by accuracy alone but by trustworthiness — combining calibration, explanation faithfulness, efficiency and out-of-distribution robustness — and that uses explainability to expose dataset shortcut learning. This is novel as a synthesis, is fully achievable with public data in one month, and produces practical model-selection guidance for real deployment.

4. Research Questions and Hypotheses

ID	Research Question	Hypothesis to test
RQ1	Which transfer-learning architecture achieves the best in-distribution classification performance?	ViT-Base and EfficientNet-B0 are competitive; DenseNet121 is a strong medical-imaging baseline.
RQ2	Do Vision Transformers outperform CNNs, and does the answer change under external (out-of-distribution) testing?	Any in-distribution ViT advantage shrinks or reverses on external data; robustness ranking differs from accuracy ranking.

ID	Research Question	Hypothesis to test
RQ3	Which architecture is best calibrated, and does temperature scaling fix over-confidence equally for CNNs and ViT?	All models are over-confident before calibration; temperature scaling reduces ECE, but by different amounts per architecture.
RQ4	Whose explanations most faithfully localise the lung pathology rather than image artefacts — and can explanations detect shortcut learning?	Models with high accuracy but poor localisation are relying on shortcuts; XAI reveals this.
RQ5	Which model offers the best overall balance of accuracy, calibration, explainability, robustness and efficiency for deployment?	No single winner on all axes; a Pareto trade-off exists and is quantifiable.

5. Research Objectives

1. Build one identical, reproducible training and preprocessing pipeline for all four architectures so comparisons are fair.
2. Benchmark ResNet50, DenseNet121, EfficientNet-B0 and ViT-Base on the primary dataset for accuracy and standard classification metrics.
3. Measure and correct confidence calibration (ECE, Brier score, reliability diagrams, temperature scaling) for every model.
4. Generate and quantitatively compare explanations (Grad-CAM for CNNs, attention rollout for ViT, SHAP where feasible), including a faithfulness/localisation score, not just qualitative pictures.
5. Test out-of-distribution robustness by evaluating each trained model on an external dataset, and use explanations to diagnose shortcut learning.
6. Benchmark computational efficiency (parameters, FLOPs, inference time, model size, GPU memory) for deployment suitability.
7. Synthesise the results into practical model-selection guidance and a camera-ready paper.

6. Expected Contributions

8. A unified, open-source benchmark comparing three CNNs and a Vision Transformer under identical preprocessing and training on chest X-rays.
9. The first (to our knowledge) five-axis trustworthiness comparison of these architectures: accuracy, calibration, explanation faithfulness, efficiency, and cross-dataset robustness together.
10. Use of explainability (Grad-CAM / attention rollout / SHAP) as a quantitative audit for shortcut learning on the COVID-19 Radiography dataset, with a localisation-based faithfulness metric.
11. Evidence on whether the global-attention bias of ViT helps or hurts out-of-distribution robustness relative to CNNs.
12. Practical, reproducible deployment recommendations depending on whether the priority is accuracy, trust, or compute budget.

7. Datasets (Availability Verified)

All datasets below are publicly available for research. The primary set is small enough to finish in a month; the external set is what makes the robustness claim possible.

Dataset	Content & size	Role in this study	Notes
COVID-19 Radiography Database (Kaggle)	4 classes: COVID, Normal, Viral Pneumonia, Lung Opacity; ~21,000 images	PRIMARY — train / val / test	Clean, balanced-ish, beginner-friendly. Known to contain source bias — which is exactly what we study.
RSNA Pneumonia / external CXR source	Pneumonia + normal frontal CXRs	EXTERNAL — out-of-distribution robustness test only (never trained on)	Different hospitals/machines; exposes shortcut learning.
NIH ChestX-ray14	>100,000 images, 14 disease labels (multi-label)	OPTIONAL — extra external generalisation check if time permits	Larger, multi-label, heavier preprocessing.
CheXpert	~220,000 images, 14 findings	OPTIONAL / future work	Very large; high storage & compute. Out of scope for 1 month.

Dataset strategy decision

Train and tune ONLY on COVID-19 Radiography. Report full metrics there.

Then evaluate every frozen model on the external RSNA-based set WITHOUT any retraining. The accuracy drop between internal and external tests is our robustness measure.

This single design choice converts a crowded 'accuracy comparison' into a publishable 'trustworthiness' study.

8. Models (Availability Verified in timm / torchvision)

All four architectures are available as ImageNet-pretrained models in the timm library and can be loaded in a single line, e.g. `tim.create_model('vit_base_patch16_224', pretrained=True, num_classes=4)`. This guarantees the modelling work is realistic within the timeline.

Model	Family	~Params	Why included
ResNet50	CNN (residual)	25.6 M	Universal, stable baseline; well understood.
DenseNet121	CNN (dense)	8.0 M	Excellent feature reuse; a favourite in medical imaging.
EfficientNet-B0	CNN (scaled)	5.3 M	Smallest & fastest; strong accuracy-per-FLOP; deployment candidate.
ViT-Base/16	Vision Transformer	86 M	Global self-attention; the CNN-vs-Transformer contrast and attention-based XAI.

Note on model size: the large parameter gap between ViT-Base (86M) and EfficientNet-B0 (5.3M) is deliberate — it lets us discuss the accuracy-vs-efficiency trade-off honestly. If ViT training proves unstable on the small dataset, a data-efficient variant (DeiT or ViT-Small) is a documented fallback.

9. Methodology

9.1 Data preprocessing

- Remove corrupted / duplicate images; verify labels.
- Resize to 224×224 ; convert grayscale to 3-channel; normalise with ImageNet mean/std.
- Stratified split 70% train / 15% validation / 15% test, fixed random seed for reproducibility.
- Training augmentation: rotation, horizontal flip, brightness/contrast jitter, zoom, shift, random crop.

- **Optional lung-segmentation branch:** run experiments on both full images and lung-segmented images (U-Net mask) to test whether removing background artefacts reduces shortcut learning — a direct, high-value experiment.

9.2 Transfer-learning strategy (identical for every model)

Phase 1 — feature extraction: freeze the pretrained backbone, train only the new classification head until the validation loss plateaus.

Phase 2 — fine-tuning: unfreeze the final blocks and fine-tune end-to-end with a lower learning rate. Compare frozen vs fine-tuned for every architecture (this is one of our controlled experiments).

Shared training controls: same optimiser (AdamW), same LR schedule family, same batch size where memory allows, same early-stopping and class-weighting, same seed — so any performance difference is attributable to architecture, not to tuning luck.

9.3 Explainability (used as an audit tool)

- **Grad-CAM** — class-discriminative heatmaps over the last convolutional layer of each CNN.
- **Attention Rollout** — aggregates ViT self-attention across layers to show where the transformer looks.
- **SHAP** — pixel/region attribution for a sample of cases (kernel/gradient explainer), for cross-method agreement.
- **Faithfulness / localisation score** — overlap between the explanation heatmap and the lung region (via the segmentation mask). Low overlap on a high-accuracy model = evidence of shortcut learning. This turns pretty pictures into a measurable number.

9.4 Confidence calibration

A model can be accurate yet dangerously over-confident. For each model we compute Expected Calibration Error (ECE) and Brier score, plot reliability diagrams, then apply temperature scaling on the validation set and re-measure. We report calibration before and after, per architecture.

9.5 Robustness / cross-dataset protocol

Each fully-trained model is evaluated, without any retraining, on the external dataset. We report the internal-to-external drop in accuracy and AUROC. We then overlay explanations on external images to visualise how the model's focus shifts when the source changes. This is the core of the trustworthiness contribution.

9.6 Efficiency benchmarking

For deployment realism we record parameter count, FLOPs, model file size, mean inference time per image (CPU and GPU), and peak GPU memory — measured under identical conditions.

10. Evaluation Metrics

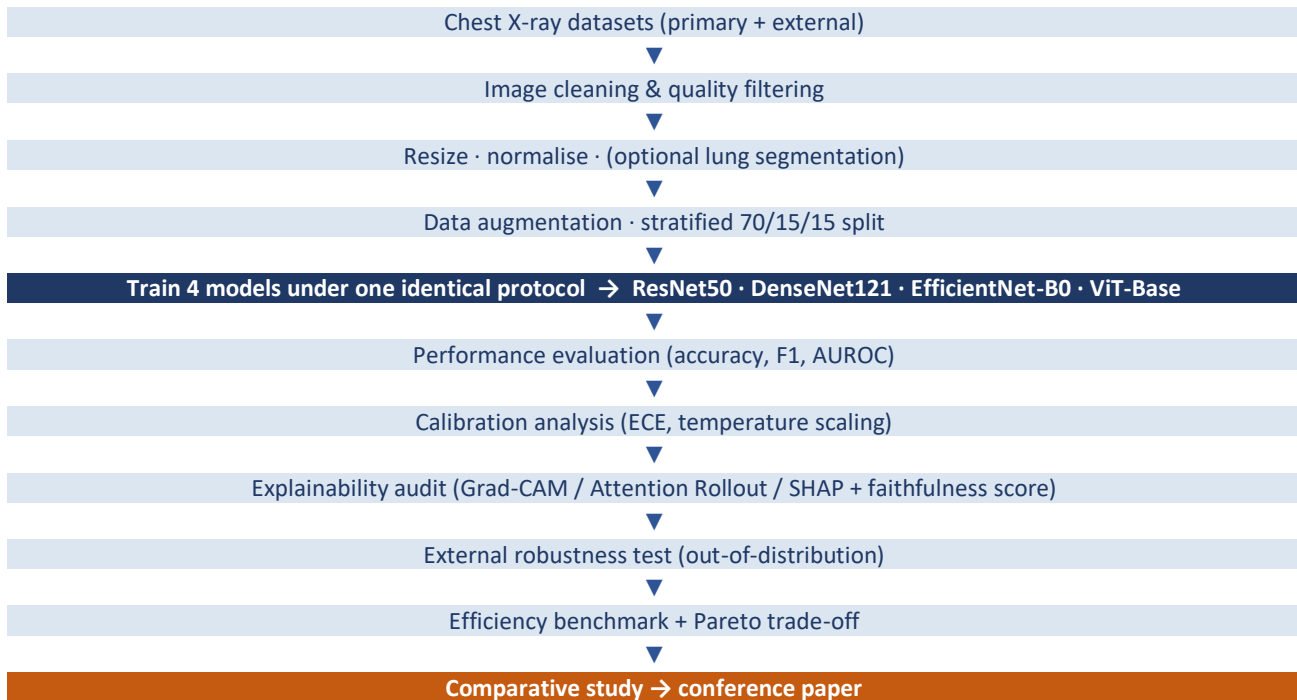
Category	Metrics
Classification	Accuracy, Precision, Recall, F1-score, ROC-AUC (per class + macro), Confusion matrix
Calibration	Expected Calibration Error (ECE), Brier score, reliability diagram (pre/post temperature scaling)
Explainability	Qualitative heatmaps + quantitative lung-localisation / faithfulness overlap score, cross-method agreement
Robustness	Internal vs external accuracy & AUROC drop (out-of-distribution generalisation gap)

Category	Metrics
Efficiency	Parameters, FLOPs, model size, inference time (CPU/GPU), peak GPU memory

11. Planned Experiments

#	Experiment	Answers
E1	Compare the three CNNs (ResNet50, DenseNet121, EfficientNet-B0) on the primary dataset.	RQ1
E2	CNNs vs Vision Transformer on the primary dataset.	RQ1, RQ2
E3	Frozen vs fine-tuned transfer learning for every model.	RQ1
E4	Full-image vs lung-segmented input (shortcut-learning mitigation).	RQ4
E5	Calibration: ECE/Brier before and after temperature scaling.	RQ3
E6	Explainability comparison + faithfulness score (Grad-CAM vs Attention Rollout vs SHAP).	RQ4
E7	External / out-of-distribution robustness test on unseen dataset.	RQ2, RQ5
E8	Efficiency benchmark + final Pareto trade-off analysis.	RQ5

12. System Pipeline



13. How to Start — Initial Steps

This section is written for a team new to deep-learning research. Do these in order; do not skip the small baseline before the big comparison.

Step 0 — Environment (Day 1)

- Create a GitHub repo. Everyone works in branches; agree on a folder structure (data/, src/, notebooks/, results/, paper/).
- Use Google Colab Pro or Kaggle Notebooks for free GPUs — no local GPU needed. Kaggle also hosts the dataset directly.
- Set up Python 3.10+, PyTorch, timm, torchvision, scikit-learn, matplotlib, grad-cam (pytorch-grad-cam), shap. Pin versions in requirements.txt.

Step 1 — Get and understand the data (Days 1–3)

- Download the COVID-19 Radiography Database from Kaggle. Count images per class; plot examples; note the class imbalance.
- Write the preprocessing + augmentation + split code once, as a shared module everyone imports.

Step 2 — One working baseline before anything else (Days 3–6)

- **Train a single model (DenseNet121) end-to-end and get a confusion matrix and accuracy.** This proves the whole pipeline works before you scale to four models. Do not parallelise the team until this baseline runs.

Step 3 — Scale to the full comparison (Week 2–3)

- Reuse the exact pipeline for the other three models — change only the model name. Log everything (TensorBoard or Weights & Biases).
- Save every checkpoint and every metric to results/ so the paper writer has data to work from.

Step 4 — Trust experiments (Week 3)

- Add calibration, explainability, and the external-dataset test. These are what make the paper novel — budget real time for them.

Step 5 — Write continuously (from Week 1)

- Member 1 starts the Introduction and Related Work in Week 1 — do not leave writing to the last week. Keep a shared reference manager (Zotero).

The single most important rule

Freeze one identical training/preprocessing pipeline early and make every model use it. Fair comparison is the whole point of the paper — if models are tuned differently, the results are not publishable.

14. Five-Member Team Division

Member	Role	Key deliverables
M1	Literature review & writing lead	Related work, research-gap section, introduction, reference management, paper coordination.

Member	Role	Key deliverables
M2	Data & preprocessing engineer	Clean dataset, augmentation + split module, shared data loaders, lung-segmentation masks.
M3	CNN modelling	ResNet50, DenseNet121, EfficientNet-B0 training + tuning, checkpoints, learning curves.
M4	Transformer & explainability	ViT-Base training, Grad-CAM, attention rollout, SHAP, faithfulness score, all XAI figures.
M5	Evaluation, calibration & robustness	Calibration + external-dataset experiments, efficiency benchmark, comparison tables, results, conclusion, slides.

Shared tasks: everyone reviews at least one section of the paper; M1 and M5 co-own the final manuscript; all code is peer-reviewed via pull requests.

15. One-Month Timeline

Week	Focus	Milestone / gate
1	Literature review, environment, data pipeline, single-model baseline.	Working baseline runs end-to-end; intro + related-work drafted.
2	Train all CNNs + ViT; frozen vs fine-tuned; log all metrics.	All four models trained; core accuracy tables complete.
3	Calibration, explainability + faithfulness, external robustness test, efficiency benchmark.	All 'trust' experiments done; figures generated.
4	Analysis, Pareto trade-off, write results/discussion/conclusion, build slides.	Camera-ready paper + presentation.

Buffer advice: ViT training and the explainability/robustness experiments are the most likely to overrun. If time is tight, drop the optional NIH ChestX-ray14 check and the SHAP analysis first — the core five-axis comparison must be protected.

16. Tools and Frameworks

Purpose	Tool
Language / DL framework	Python · PyTorch
Pretrained models & transfer learning	timm · torchvision
Explainability	pytorch-grad-cam · attention rollout · SHAP
Calibration	custom ECE/Brier + temperature scaling
Visualisation	Matplotlib · Seaborn
Experiment tracking	Weights & Biases or TensorBoard
Compute	Google Colab Pro / Kaggle GPUs
Version control & writing	Git + GitHub · Zotero · Overleaf (LaTeX) or Word

17. Risks and Mitigations

Risk	Mitigation
ViT under-performs / over-fits on the small dataset.	Use strong augmentation and Phase-1 freezing; fall back to DeiT / ViT-Small (data-efficient).
External dataset formats/labels differ.	Map to a shared label space (COVID/pneumonia/normal); pick the external set in Week 1, not Week 3.

Risk	Mitigation
Compute limits (free GPU quotas).	224×224 inputs, mixed-precision training, checkpoint frequently, stagger jobs across accounts.
SHAP too slow on images.	Run SHAP only on a small representative sample; it is optional, not core.
Scope creep / running out of time.	Protect the five-axis core; NIH check and SHAP are the first things to cut.

18. Expected Outcomes and Target Venues

Deliverables: (1) an open-source, reproducible benchmark repository; (2) a set of comparison tables and explainability/robustness figures; (3) a full paper with practical model-selection guidance; (4) a presentation.

Suitable venues include undergraduate / student research conferences, medical-imaging and applied-AI workshops, and open-access journals or preprint servers (arXiv, then a journal such as a Scientific Reports-style venue or an IEEE/Springer conference). The reproducibility and trustworthiness framing fit these audiences well and are realistic for a first paper.

19. Indicative References (to be expanded by M1)

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- [10] Enhanced tuberculosis detection using Vision Transformers and explainable AI with a Grad-CAM approach on chest X-rays. BMC Medical Imaging (2025).
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